

Methods of Theoretical Physics

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Sets

Lecture 1: Syllabus Day

Tue 21 Jan 2025 12:30

The professor is recommending ChatGPT as a tutor. This is awesome. Problem sets due Thursdays at 11:59pm.

Prof asked “What are numbers?” and this chick answered “erm uh numebrs are uh do you want me to give the set theory definition? Erm its difficult to define numbers it requires complex set theory” lmfaol. We will define numbers as an element of an algebraic structure, like a field or group. We take $0 \in \mathbb{N}$. We use the convention of writing $\mathbb{Z}_2$ instead of $\mathbb{Z}/2\mathbb{Z}$.

A cartesian product is a product in **Set**; an object $A \times B$ together with canonical projections that are final in the category $\mathbf{Set}^{A,B}$.